

Workbook #1

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Please submit your progress report using the (updated) [Workbook1 template](https://sjsu.instructure.com/courses/1595447/files/78149854?wrap=1) (<https://sjsu.instructure.com/courses/1595447/files/78149854?wrap=1>). Name the file as **Team#-Workbook1.doc** where # is the team number. Here's a [sample Chapter 5](https://sjsu.instructure.com/courses/1595447/files/78149843?wrap=1) (<https://sjsu.instructure.com/courses/1595447/files/78149843?wrap=1>) for your reference.

Here's a [recorded lecture](https://sjsu.instructure.com/courses/1595447/pages/chapter-5-lecture) (<https://sjsu.instructure.com/courses/1595447/pages/chapter-5-lecture>) from Spring 2022 that covers some of the details for this workbook and talks about **system boundary** in details.

298B Workbook 1 Rubrics

Criteria	Ratings			Pts
4.1 Model Proposals Propose models to address targeted problems in detailed terms of concepts, features, model architectures, algorithms, etc.	0.5 to >0.4 pts Excellent Propose 5 or more models and adequately their address targeted problems in detailed terms of concepts, features, model architectures, algorithms, etc. At least one model is innovative or improved compared to the existing models.	0.4 to >0.2 pts Good Propose 3 or more models and adequately address their targeted problems in detailed terms of concepts, features, model architectures, algorithms, etc.	0.2 to >0 pts Poor Propose less than 3 models or inadequately address their targeted problems in detailed terms of concepts, features, model architectures, algorithms, etc.	0.5 pts
4.2. Model Supports (1) Describe the platform, environment and tools to support the development and execution of each model. (2) Provide diagrams of supporting platform architecture, components, machine learning data flows, etc.	0.5 to >0.4 pts Excellent (1) Correctly and clearly describe the platform, environment and tools to support the development and execution of each model. (2) Provide sufficient and excellent diagrams of supporting platform architecture, components, machine learning data flows, etc.	0.4 to >0.2 pts Good (1) Correctly describe the platform, environment and tools to support the development and execution of each model. (2) Provide sufficient diagrams of supporting platform architecture, components, machine learning data flows, etc.	0.2 to >0 pts Poor (1) Incorrectly describe the platform, environment and tools to support the development and execution of each model, or (2) Provide insufficient or inaccurate diagrams of supporting platform architecture, components, machine learning data flows, etc.	0.5 pts

Criteria	Ratings			Pts
4.3 Model Comparison and Justification (1) Compare models including targeted problems, features, approaches (statistics, machine learning, and deep learning), strengths and limitations. (2) Provide justifications for each model.	1 to >0.8 pts Excellent (1) Correctly and thoroughly compare models including targeted problems, features, approaches (statistics, machine learning, and deep learning), strengths and limitations in a good format. (2) Provide sufficient justifications for each model.	0.8 to >0.5 pts Good ((1) Correctly and adequately compare models including targeted problems, features, approaches (statistics, machine learning, and deep learning), strengths and limitations. (2) Provide sufficient justifications for each model.	0.5 to >0 pts Poor (1) Incorrectly or insufficiently compare models including targeted problems, features, approaches (statistics, machine learning, and deep learning), strengths and limitations, or (2) Lack of justifications for each model.	1 pts
4.4 Model Evaluation Method Present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc.	1 to >0.8 pts Excellent Correctly, completely and clearly present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc.	0.8 to >0.5 pts Good Correctly and completely present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc.	0.5 to >0 pts Poor Either incorrectly or incompletely present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc.	1 pts
5.1 System Requirements Analysis (1) Identify system boundary, actors and use cases. (2) Describe high-level data analytics and machine learning functions and capabilities.	2 to >1.6 pts Excellent (1) Complete, correct and clear identification of system boundary, actors and use cases. (2) Complete, correct and clear description of high-level data analytics and machine learning functions and capabilities.	1.6 to >1.0 pts Good (1) Complete and correct identification of system boundary, actors and use cases. (2) Complete and correct description of high-level data analytics and machine learning functions and capabilities.	1 to >0 pts Poor (1) Incomplete or incorrect identification of system boundary, actors and use cases. (2) incomplete or incorrect description of high-level data analytics and machine learning functions and capabilities.	2 pts
5.2 System Design (1) Design the system	2 to >1.6 pts Excellent	1.6 to >1.0 pts Good	1 to >0 pts Poor	2 pts

Criteria	Ratings			Pts
architecture and infrastructure with AI-powered function components. (2) Design the integration with supporting platforms, frameworks and a cloud environment. (3) Design the data management solution and database. (4) Design user interface and data visualization.	(1) Complete and correct design of an excellent system architecture and infrastructure with AI-powered function components. (2) Complete and correct design of integration with supporting platforms, frameworks and a cloud environment. (3) Complete and correct design of data management solution and database. (4) Complete and correct design of excellent user interface and data visualization.	(1) Complete and correct design of a system architecture and infrastructure with AI-powered function components. (2) Complete and correct design of integration with supporting platforms, frameworks and a cloud environment. (3) Complete and correct design of data management solution and database. (4) Complete and correct design of excellent user interface and data visualization.	(1) Incomplete or incorrect design of a system architecture and infrastructure with AI-powered function components. (2) Incomplete or incorrect design of integration with supporting platforms, frameworks and a cloud environment. (3) Incomplete or incorrect design of data management solution and database. (4) Incomplete or incorrect design of excellent user interface and data visualization.	
5.3 Intelligent Solution (1) Present developed AI and machine learning solutions for each targeted problem, including integrated solutions, ensembled, developed and applied machine learning models. (2) Describe required project input datasets, expected outputs, supporting system contexts, and solution APIs.	2 to >1.6 pts Excellent (1) Complete and correct development of AI and machine learning system including proposed, ensembled, developed and applied machine learning models. (2) Complete, correct and clear description of the required project input datasets, expected outputs, supporting system contexts, and solution APIs.	1.6 to >1.0 pts Good (1) Complete and correct development of AI and machine learning system including proposed, ensembled, developed and applied machine learning models. (2) Complete and correct description of the required project input datasets, expected outputs, supporting system contexts, and solution APIs.	1 to >0 pts Poor (1) Incomplete or incorrect development of AI and machine learning system including proposed, ensembled, developed and applied machine learning models. (2) Incomplete or incorrect description of the required project input datasets, expected outputs, supporting system contexts, and solution APIs.	2 pts
5.4. System Support Environment	1 to >0.8 pts Excellent	0.8 to >0.5 pts Good	0.5 to >0 pts Poor	1 pts

Criteria	Ratings			Pts
Present the information and features of system supporting environment, including technologies, platforms, frameworks, etc.	Complete, correct, and clear presentation regarding the information and features of system supporting environment.	Complete and correct presentation regarding the information and features of system supporting environment.	Incomplete or incorrect presentation regarding the information and features of system supporting environment.	
				Total Points: 10